

From Assistance to Authorship

A Framework for Evaluating AI-Assisted Academic Work in Higher Education

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This framework is provided to support scholarly discussion, institutional implementation, and professional practice. It does not constitute institutional policy or regulatory guidance unless formally adopted by the relevant institution.

About this Framework

Generative artificial intelligence has transformed academic work more rapidly than institutional policies have been able to adapt. Across higher education, educators are increasingly confronted with a question that extends beyond whether AI has been used:

When does AI-assisted work remain genuinely the work of the human author?

This framework addresses that question by shifting attention away from technology itself and towards the conditions that preserve legitimate academic authorship. Rather than evaluating AI as inherently acceptable or unacceptable, it evaluates whether the human author's intellectual contribution remains central throughout the development of academic work.

The framework proposes that AI-assisted academic work remains legitimate only when three interdependent principles are simultaneously upheld:

- **Epistemic Responsibility** — the human author retains intellectual ownership of ideas, arguments, interpretations, analytical decisions, and conclusions.
- **Functional Limitation** — AI performs supportive rather than substitutive functions and does not replace disciplinary reasoning or scholarly judgement.
- **Transparency and Disclosure** — AI involvement is communicated appropriately to enable informed academic evaluation and accountability.

These principles operate as an integrated evaluative framework rather than as independent requirements. None is sufficient in isolation. Disclosure cannot legitimise inappropriate AI use, responsibility cannot compensate for concealed assistance, and limiting AI's role without preserving intellectual ownership does not maintain authentic authorship. Academic legitimacy therefore emerges from the interaction of all three principles.

The framework is intended for educators, researchers, programme leaders, assessment designers, academic integrity professionals, quality assurance teams, and institutional policy developers seeking a principled approach to evaluating AI-assisted academic work. It also serves as the conceptual foundation for a broader suite of companion frameworks that translate these principles into practical guidance for assessment design, disclosure, verification, institutional governance, and educational practice.

Executive Summary

Generative artificial intelligence is reshaping academic writing, assessment, and knowledge production. Universities increasingly recognise that traditional distinctions between original authorship and external assistance are becoming more difficult to apply in environments where AI systems can generate coherent academic text, propose arguments, organise evidence, and imitate disciplinary writing styles. Existing institutional responses have largely focused on detection technologies, disclosure requirements, or lists of permitted and prohibited practices. While useful, these approaches often fail to

explain why some forms of AI assistance remain academically legitimate whereas others undermine authorship.

The present framework proposes a different starting point. Instead of evaluating AI itself, it evaluates whether human authorship has been preserved.

The framework identifies three interacting principles that determine the legitimacy of AI-assisted academic work.

Epistemic Responsibility requires that the human author remains intellectually accountable for the ideas, interpretations, arguments, analytical decisions, and conclusions presented within the work. AI may assist expression, but it cannot assume responsibility for scholarly judgement.

Functional Limitation defines the appropriate role of AI within academic work. AI may support planning, drafting, language refinement, organisation, and other subsidiary activities, but it should not replace disciplinary reasoning, generate original scholarly claims on behalf of the author, or determine the intellectual direction of the work.

Transparency and Disclosure require that AI involvement is communicated honestly and proportionately. Disclosure allows readers, supervisors, reviewers, and assessors to evaluate the nature and extent of AI assistance. Transparency strengthens accountability but does not independently legitimise inappropriate use.

The framework therefore conceptualises academic legitimacy as a conditional outcome emerging from the interaction of responsibility, limitation, and disclosure rather than from the presence or absence of AI itself. This shifts institutional attention from detecting technology towards evaluating human intellectual engagement.

The framework is intended to support multiple areas of higher education practice, including assessment design, supervision, academic writing, faculty development, institutional policy, moderation, academic integrity processes, and AI governance. It also provides the conceptual foundation for six companion implementation resources that operationalise its principles through disclosure guidance, assessment design tools, verification approaches, governance processes, and institutional implementation models.

This publication is **conceptual and practice-informed**. It draws upon existing scholarship, professional practice, and ongoing research into AI-assisted academic writing, including ongoing exploratory empirical research examining how experienced academics evaluate the legitimacy of AI-assisted writing. It should not be interpreted as a validated measurement instrument or universal regulatory model. Rather, it offers a structured framework to support scholarly discussion, institutional adaptation, and responsible

educational practice as higher education continues to respond to rapidly evolving AI technologies.

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1. Introduction

Generative artificial intelligence has altered academic practice with unprecedented speed. Within a few years, systems capable of producing fluent academic prose, summarising research, generating literature reviews, proposing arguments, explaining concepts, and responding to complex prompts have become readily accessible to students, researchers, and educators. Their widespread adoption has prompted universities worldwide to reconsider long-established assumptions about authorship, originality, assessment, and academic integrity.

Much of the early institutional response focused on identifying AI-generated text. Detection software, policy statements, and restrictions on AI use became common features of higher education. As experience accumulated, however, it became increasingly clear that the central challenge was not technological detection. AI systems could assist learning in ways that were educationally valuable while also enabling practices that undermined genuine intellectual engagement. Between these extremes lay a broad spectrum of legitimate, questionable, and context-dependent uses that could not be addressed adequately through binary categories of acceptable or unacceptable AI use.

The question facing higher education has therefore evolved. Rather than asking whether AI has been used, educators are increasingly asking under what conditions AI-assisted work remains authentically the work of the student or scholar who submits it.

The proposed model responds to that question by shifting attention from technology to authorship. It argues that the legitimacy of AI-assisted academic work depends not on the presence of AI itself, but on whether the human author continues to exercise meaningful intellectual responsibility, whether AI remains confined to appropriate supportive functions, and whether its contribution is communicated transparently. These three conditions—responsibility, functional limitation, and disclosure—provide a principled basis for evaluating AI-assisted work across diverse educational contexts while preserving the core values of scholarship.

Unlike many institutional guidance documents, the proposed framework is not intended to prescribe detailed rules for every possible use of AI. The diversity of disciplines, assessment formats, educational purposes, and technological developments makes exhaustive regulation both impractical and undesirable. Instead, the framework offers a stable conceptual foundation that can guide context-sensitive professional judgement. It seeks to support educators, researchers, policy makers, and institutions in making consistent decisions while recognising that legitimate academic practice will continue to evolve alongside AI technologies.

The framework also reflects a broader shift in higher education. Academic integrity is increasingly understood not simply as the prevention of misconduct but as the cultivation of responsible scholarship, authentic learning, and intellectual accountability. In this context, AI should not be viewed solely as a source of academic risk. Properly integrated, it has the potential to support learning, improve accessibility, enhance productivity, and strengthen educational practice. The challenge is ensuring that these benefits are realised without weakening the intellectual responsibilities that define academic authorship.

This publication therefore proposes a framework that is simultaneously normative and practical. It establishes conceptual principles for evaluating AI-assisted academic work while providing the foundation for a broader suite of implementation resources designed to support responsible AI governance throughout higher education.

2. Why a New Framework is Needed

The Shift from AI Detection to Academic Judgement

The arrival of generative artificial intelligence has disrupted higher education in ways few technologies have done before. Earlier digital innovations changed *how* students accessed information, communicated, or submitted work. Generative AI changes something more fundamental: it participates directly in the production of academic text. This development challenges one of the central assumptions underpinning higher education—that submitted work represents the intellectual activity of the named author.

Initial institutional responses reflected understandable concern. Universities rapidly revised academic integrity policies, introduced guidance on acceptable AI use, experimented with AI-detection software, and developed disclosure requirements. Much of this activity was necessarily reactive. Institutions sought to preserve assessment integrity while responding to technologies that were evolving faster than policy could reasonably accommodate.

Yet experience over the past several years has exposed important limitations in these early responses. Detection technologies have proved unreliable, institutional policies vary considerably, and educators continue to face difficult judgements that cannot be resolved simply by determining whether AI has been used. The practical question encountered in marking, supervision, peer review, and research assessment is rarely **"Was AI involved?"** Instead, it is more often **"Does this work still represent the author's own intellectual contribution?"**

This distinction is crucial. AI use is no longer exceptional. Students routinely use AI to clarify concepts, generate practice questions, organise notes, improve language, translate ideas across languages, and obtain feedback on drafts. Researchers increasingly employ AI to refine prose, explore alternative structures, summarise literature, or improve readability. Treating every instance of AI use as academically problematic is neither realistic nor educationally desirable. Equally, treating all AI assistance as benign ignores genuine concerns regarding authorship, accountability, and scholarly responsibility.

Higher education therefore requires a more sophisticated evaluative approach.

The challenge is no longer to distinguish between work produced *with* AI and work produced *without* AI. That distinction is becoming increasingly artificial as AI becomes integrated into everyday academic practice. Instead, institutions must distinguish between AI that **supports** human intellectual work and AI that **substitutes** for it. The boundary between these two forms of assistance cannot be determined solely by software, policy checklists, or disclosure statements. It requires informed academic judgement.

This model begins from that premise.

Rather than asking whether AI is present, it asks whether the defining characteristics of academic authorship remain visible. It assumes that authorship is not simply a matter of producing text but of exercising intellectual responsibility for what the text claims, argues, and concludes. Consequently, legitimacy depends not on technological purity but on preserving the relationship between the human author and the intellectual work presented.

This shift has important implications. It moves institutional attention away from technology itself and towards the educational values that technology may either support or undermine. It also recognises that similar AI tools can be used in fundamentally different ways. A student who uses AI to improve grammatical accuracy while independently constructing an argument presents a different educational case from one who asks AI to generate the argument itself. Both have used AI; only one has delegated the intellectual task that the assessment was designed to evaluate.

The framework therefore rejects binary classifications of AI use as simply acceptable or unacceptable. Academic practice rarely operates through such rigid distinctions. Instead, educators routinely exercise professional judgement by considering context, purpose, disciplinary expectations, assessment design, and evidence of student understanding. AI-assisted academic work should be evaluated in the same way: through principled judgement grounded in clear evaluative criteria rather than mechanical rules.

Such an approach is also more sustainable. AI technologies will continue to evolve, making detailed lists of permitted and prohibited activities increasingly difficult to maintain. A

framework built around enduring academic principles is more likely to remain useful than guidance tied to the capabilities of specific tools or models. Responsibility, scholarly judgement, intellectual ownership, and transparency are longstanding academic values. While technologies will change, these principles remain central to what universities recognise as legitimate academic work.

For this reason, the framework is intentionally technology-neutral. It is not a framework for evaluating ChatGPT, Claude, Gemini, or any other particular system. It is a framework for evaluating human academic practice in environments where AI has become an ordinary part of scholarly work. Its concern is therefore not artificial intelligence itself, but the preservation of responsible authorship within increasingly AI-mediated educational contexts.

3. The Challenge of AI-Assisted Authorship

Beyond Originality

For generations, academic integrity has been closely associated with originality. Students were expected to produce their own work, acknowledge the ideas of others, and avoid practices such as plagiarism, collusion, and contract cheating. Although these principles remain important, generative AI reveals that originality alone is no longer sufficient as the organising concept for academic evaluation.

A student may submit entirely original text that nevertheless reflects little personal intellectual engagement because the underlying ideas, analytical structure, and interpretation have been generated by AI. Conversely, another student may legitimately use AI to improve expression or overcome linguistic barriers while retaining complete ownership of the conceptual work. Judging these two cases through originality alone obscures the more fundamental educational issue.

The central question is not whether words are new. It is whether the intellectual work represented by those words genuinely belongs to the named author.

This distinction requires a reconsideration of authorship itself. Traditionally, authorship has implied both ownership and responsibility. Authors receive recognition for their contributions because they also accept accountability for the accuracy, coherence, interpretation, and implications of what they publish. AI complicates this relationship by enabling sophisticated textual production without corresponding intellectual engagement. Fluent prose can now be generated independently of disciplinary understanding, critical evaluation, or scholarly judgement.

The result is a growing separation between textual production and intellectual responsibility. Academic writing can appear authoritative while masking uncertainty about who—or what—performed the underlying intellectual work.

The purpose of this model is not to restore a pre-AI conception of authorship, nor to argue that academic work should remain entirely unaided by technology. Such a position would ignore the long history of technological assistance within higher education, from spelling and grammar software to reference managers, translation tools, statistical packages, and collaborative writing platforms. Universities have consistently accommodated technologies that support academic work without replacing the intellectual responsibilities that define scholarship.

Generative AI differs not because it is digital, but because it can contribute directly to activities traditionally regarded as evidence of human scholarship: explanation, synthesis, interpretation, evaluation, and argumentation. These are precisely the forms of intellectual work through which universities assess learning, recognise expertise, and confer academic qualifications.

The challenge therefore is not technological but educational. Higher education must determine where appropriate assistance ends and intellectual substitution begins. This boundary cannot be established solely by counting words generated by AI or requiring disclosure statements. It depends on whether the author continues to exercise meaningful scholarly judgement throughout the development of the work.

For this reason, the framework proposed in this publication places authorship—not AI—at the centre of academic evaluation. AI is the context. Authorship remains the object of evaluation.

4. The Three Foundational Principles

Overview

The framework proposed in this publication rests upon three interdependent principles that together determine the legitimacy of AI-assisted academic work:

- **Epistemic Responsibility**
- **Functional Limitation**
- **Transparency and Disclosure**

These principles should not be interpreted as independent criteria against which academic work is assessed separately. Their significance lies in their interaction. A piece of work may satisfy one or even two principles while still failing to demonstrate legitimate authorship. Likewise, no single principle carries sufficient weight to determine legitimacy in isolation.

For example, a student may disclose extensive AI use honestly. Such transparency is ethically preferable to concealment, but disclosure alone does not establish that the intellectual work remains the student's own. Similarly, a student may genuinely understand a topic and therefore retain epistemic responsibility, yet still permit AI to perform analytical tasks that the assessment was explicitly designed to evaluate. Conversely, AI may be confined to relatively minor linguistic support, but undisclosed use may undermine institutional expectations of openness where disclosure is required.

The framework therefore treats legitimacy as a relational judgement emerging from the interaction of all three principles. Their combined operation preserves the essential relationship between academic work and the human author who claims ownership of it.

This relational perspective also reflects how educators actually evaluate academic work. Experienced supervisors, markers, reviewers, and examiners rarely rely upon single indicators when judging authenticity or academic integrity. Instead, they consider multiple forms of evidence simultaneously, balancing intellectual ownership, disciplinary expectations, contextual information, and professional judgement. The framework seeks to make that evaluative reasoning more explicit.

4.1 Epistemic Responsibility

The Intellectual Core of Authorship

Among the three principles, epistemic responsibility occupies a foundational position. Without it, authorship loses its defining characteristic. Academic writing has never been valued simply because it contains information. Its educational and scholarly significance derives from the expectation that the named author accepts responsibility for the knowledge claims presented within the work.

Responsibility, in this context, extends well beyond factual accuracy. It encompasses the intellectual decisions that shape scholarly inquiry: selecting evidence, interpreting findings, constructing arguments, evaluating competing perspectives, defining conceptual boundaries, acknowledging uncertainty, and drawing conclusions. These activities require judgement rather than linguistic fluency. They constitute the intellectual labour that universities seek to cultivate and assess.

Generative AI complicates this relationship because it can produce text that resembles the products of scholarly judgement without participating in judgement itself. Language models predict plausible linguistic sequences based upon statistical relationships within training data. They do not understand the conceptual validity of arguments, appreciate disciplinary debates, or assume responsibility for the consequences of the claims they generate. Responsibility therefore cannot be delegated to AI because AI possesses neither accountability nor intentionality.

The implication is straightforward but important. Regardless of how extensively AI contributes to drafting or language production, responsibility for every substantive academic claim must remain attributable to the human author. Readers, supervisors, examiners, journal editors, and institutions require a clearly identifiable individual who accepts ownership of the work's intellectual content. AI cannot fulfil this role.

Accordingly, this conceptual defines **epistemic responsibility** as:

The continuing obligation of the human author to exercise intellectual ownership over the ideas, arguments, interpretations, analytical decisions, and conclusions presented within academic work, regardless of any technological assistance received during its production.

This definition deliberately shifts attention away from textual production and towards intellectual agency. The question is not who typed the words but who generated, evaluated, defended, and ultimately accepts responsibility for the thinking those words express.

Responsibility Is More Than Understanding

Discussions surrounding AI-assisted assessment often reduce authorship to a relatively simple question: *Can the student explain the work?* Oral verification interviews, viva examinations, and reflective commentaries increasingly attempt to establish authenticity by asking students to explain their reasoning after submission.

Although these practices provide valuable evidence, they should not be confused with epistemic responsibility itself.

A student may understand an AI-generated explanation perfectly without having developed the underlying reasoning independently. Likewise, an experienced researcher may competently defend a sophisticated argument originally proposed by AI after reviewing and accepting it. Understanding demonstrates comprehension; it does not necessarily demonstrate authorship.

Epistemic responsibility therefore requires more than retrospective explanation. It concerns the author's active participation throughout the intellectual development of the

work. The responsible author does not merely recognise good ideas after they appear; they make substantive decisions about which ideas to adopt, reject, modify, integrate, or challenge during the process of scholarly inquiry.

This distinction becomes increasingly important as AI systems become more capable of generating persuasive analytical content. Educational evaluation cannot rest solely upon the ability to explain finished work. It must also consider whether meaningful intellectual judgement remained continuously exercised during its creation.

Intellectual Ownership Is Distributed Across the Writing Process

Academic writing is often mistakenly viewed as the production of a final document. In reality, scholarly writing represents the culmination of numerous intellectual decisions made throughout the research and writing process.

These include decisions such as:

- identifying worthwhile questions;
- selecting theoretical perspectives;
- interpreting evidence;
- weighing alternative explanations;
- determining argumentative structure;
- recognising limitations;
- anticipating counterarguments;
- deciding what should remain uncertain.

These decisions collectively establish intellectual ownership. They cannot be reduced to the production of polished sentences.

Consequently, the framework does not ask whether AI generated particular paragraphs. Instead, it asks whether AI assumed responsibility for decisions that properly belong to the human scholar.

This distinction helps explain why similar AI tools may support legitimate learning in one context while undermining it in another. AI-generated feedback on sentence clarity may leave intellectual ownership entirely untouched. AI-generated interpretation of research findings may directly replace the reasoning the assessment was intended to evaluate.

The educational significance therefore lies not in **how much** AI produced, but **what kind of intellectual work** AI performed.

Responsibility as the Foundation of Academic Trust

Higher education ultimately depends upon trust.

Universities certify graduates because society trusts that awarded qualifications represent genuine achievement. Journals publish research because readers trust that named authors stand behind published claims. Supervisors endorse theses because they trust that doctoral candidates genuinely conducted the intellectual work described within them.

This trust rests upon identifiable human responsibility.

When responsibility becomes ambiguous, academic credibility weakens irrespective of whether AI-generated text is factually accurate. A technically flawless literature review produced largely through AI raises questions not because it necessarily contains errors, but because uncertainty remains concerning whose scholarly judgement the reader is being asked to trust.

The preservation of epistemic responsibility therefore serves purposes extending well beyond assessment integrity. It protects the credibility of academic qualifications, scholarly communication, professional expertise, and institutional reputation. In this sense, responsibility functions not simply as an educational principle but as one of the fundamental conditions upon which higher education itself depends.

4.2 Functional Limitation

Defining the Appropriate Role of Artificial Intelligence

Generative AI presents higher education with a challenge that earlier educational technologies did not. Previous tools—spell checkers, grammar assistants, search engines, reference managers, statistical software, and translation applications—extended human capability while leaving intellectual responsibility largely untouched. They improved efficiency but did not normally assume responsibility for thinking.

Generative AI is fundamentally different.

Large language models can propose research questions, generate explanations, construct arguments, synthesise literature, interpret findings, recommend conclusions, and imitate disciplinary writing with remarkable fluency. Many of these activities have traditionally

been regarded not as peripheral aspects of scholarship but as evidence of scholarly expertise itself.

Consequently, the educational question is no longer whether AI assists academic work. Assistance has always existed. The more important question is whether AI remains a **supporting tool** or becomes an **intellectual substitute**.

This proposed framework addresses that question through the principle of **functional limitation**.

Functional limitation refers to the deliberate restriction of AI to roles that support rather than replace the intellectual activities that higher education seeks to develop, evaluate, and certify. It is concerned with the *function* AI performs within the academic process rather than with the quantity of AI-generated text appearing in the final submission.

Accordingly, the framework defines functional limitation as:

The principle that AI should remain confined to supportive functions that enhance human academic work without replacing the intellectual judgement, disciplinary reasoning, analytical interpretation, or scholarly decision-making that constitute authorship.

This definition deliberately avoids prescribing specific technologies or software. The legitimacy of AI use depends not upon which application is employed but upon the nature of the contribution it makes.

Assistance and Substitution

The distinction between assistance and substitution lies at the heart of the framework.

Assistance occurs when AI enables the author to express, organise, or communicate ideas that remain fundamentally their own. The technology enhances the efficiency or clarity of scholarly work without determining its intellectual substance.

Substitution occurs when AI assumes responsibility for intellectual activities that the learner or researcher is expected to perform independently. In these situations, AI no longer supports scholarship; it performs scholarship on behalf of the author.

Although the distinction appears straightforward conceptually, it often becomes blurred in practice. Consider the following examples.

A student asks AI to improve the grammar and coherence of a discussion section that they have already written. The ideas, interpretation, and conclusions remain their own. AI contributes to expression but not to intellectual development. This represents assistance.

A second student uploads raw research findings and instructs AI to identify themes, explain their significance, compare them with previous studies, and write the discussion chapter. Although the student may subsequently review the text carefully, the central intellectual work has already been undertaken by AI. Here AI has substituted for the reasoning that the assessment was intended to evaluate.

The difference lies not in the sophistication of the software nor in the number of AI-generated words, but in the nature of the intellectual activity delegated.

The Functional Boundary

Rather than categorising AI tools as acceptable or unacceptable, the framework establishes a functional boundary between **supportive** and **substitutive** activities.

Supportive activities assist the author without displacing intellectual ownership. These commonly include:

- improving grammar, spelling, punctuation, and style;
- enhancing readability and clarity;
- suggesting alternative wording;
- formatting references and citations;
- organising notes created by the author;
- generating study questions for revision;
- explaining unfamiliar concepts for learning purposes;
- translating text while preserving the author's intended meaning;
- supporting accessibility requirements.

These functions primarily improve communication rather than generate scholarly judgement.

Substitutive activities, by contrast, involve AI performing intellectual work that should remain attributable to the human author. Examples include:

- generating original analytical arguments;
- interpreting empirical findings without independent human analysis;
- determining theoretical positioning;
- evaluating competing explanations on behalf of the author;

- drawing substantive research conclusions;
- constructing critical discussions that the author cannot independently justify;
- synthesising literature without meaningful human evaluation;
- producing assessment responses intended to demonstrate the learner's own understanding.

These activities directly influence the intellectual content of academic work. When delegated to AI, they weaken the relationship between authorship and responsibility that higher education depends upon.

Functional Limitation Is Context Dependent

The boundary between supportive and substitutive AI use cannot be reduced to universal rules. The educational purpose of the task must always be considered.

For example, AI-assisted translation may be entirely appropriate within a history assignment where historical reasoning is being assessed. The same translation assistance may be inappropriate within an assessment designed specifically to evaluate second-language writing ability.

Similarly, AI-generated coding assistance may represent an acceptable professional practice within software development while being incompatible with an introductory programming assessment intended to evaluate foundational coding skills.

The educational function of an assessment therefore determines where the functional boundary should be drawn.

This contextual perspective prevents institutions from adopting unnecessarily restrictive policies while preserving the integrity of learning outcomes. It also aligns AI governance with a longstanding principle of assessment design: students should receive support for activities that are not themselves being assessed, while independently demonstrating competence in those that are.

Functional Limitation Protects Learning Rather Than Restricting Technology

Discussions surrounding AI governance often present institutional guidance as a struggle between innovation and regulation. Such framing is misleading.

The purpose of functional limitation is not to restrict technological progress. It is to preserve educational purpose.

Universities do not assess students simply to produce documents. They assess students to determine whether graduates have developed disciplinary understanding, critical thinking, professional judgement, and independent reasoning. If AI routinely performs these intellectual activities, assessment gradually shifts from evaluating learning to evaluating effective prompting.

This distinction has profound implications for curriculum design. As AI capabilities continue to expand, universities will need to reconsider which intellectual activities should remain central to human learning and which may appropriately be supported through technology. Functional limitation provides a principled basis for making these decisions without assuming that every technological advance should automatically become educationally acceptable.

The framework therefore treats AI not as a threat to higher education but as a catalyst requiring greater clarity about the educational purposes that assessment is intended to serve. Limiting AI's role is ultimately about protecting meaningful learning rather than limiting innovation.

Relationship with Epistemic Responsibility

Functional limitation and epistemic responsibility reinforce one another.

When AI remains within supportive functions, the human author is better positioned to retain intellectual ownership of the work. Conversely, as AI assumes increasingly substantive analytical roles, responsibility becomes progressively more difficult to locate. Eventually, authorship itself becomes ambiguous.

For this reason, neither principle can operate independently. Restricting AI to supportive activities helps preserve responsibility, while responsibility provides the normative justification for limiting AI's function. Together they establish the conceptual boundary between legitimate assistance and intellectual substitution.

Running Case Study

Emily's Dissertation

Throughout this conceptual, consider the following example.

Emily is a final-year undergraduate student completing a dissertation investigating the impact of artificial intelligence on higher education. She conducts the literature review, collects and analyses her own interview data, and develops the overall argument independently.

During the writing process she uses a generative AI system in several ways:

- to improve grammar and sentence fluency;
- to reorganise awkward paragraphs;
- to suggest alternative section headings;
- to explain unfamiliar statistical terminology;
- to produce summaries of several research articles;
- to propose possible interpretations of her interview findings;
- to draft a discussion section comparing her findings with previous research.

Emily reviews every AI-generated output carefully. She revises much of the generated text, rejects some suggestions, and incorporates others into the dissertation. Her university requires disclosure of significant AI assistance but does not define precisely what "significant" means.

At first glance, Emily appears either to have used AI responsibly or to have crossed important academic boundaries. Reasonable educators may initially disagree. The purpose of this proposed model is not to produce an automatic answer but to provide a structured way of evaluating where legitimate assistance ends and intellectual substitution begins.

Each of the following chapters will revisit Emily's dissertation from a different perspective.

4.3 Transparency and Disclosure

Visibility Rather Than Permission

Among the three principles, transparency is often the most visible because it appears in institutional policies, publisher guidance, and assessment regulations. Many universities now require students to declare AI use, while publishers increasingly expect authors to acknowledge AI-assisted writing during manuscript preparation.

Disclosure has therefore become one of the most recognisable institutional responses to generative AI.

Yet transparency is also the principle most frequently misunderstood.

Disclosure is sometimes interpreted as a form of permission: provided AI use is declared, the work is assumed to remain acceptable. Others adopt the opposite position, treating disclosure as evidence that academic misconduct has occurred. Both interpretations oversimplify the role that transparency plays within responsible academic practice.

This model adopts a different position.

Transparency does not determine legitimacy.

Transparency enables legitimacy to be evaluated.

Disclosure is therefore not the end of academic judgement but its beginning.

Why Transparency Matters

Academic work depends upon informed evaluation.

Markers assess assignments.

Supervisors evaluate dissertations.

Peer reviewers assess manuscripts.

Editors determine publication decisions.

Professional bodies certify competence.

None of these activities can be performed fairly if important aspects of the academic process remain concealed.

Disclosure therefore serves an informational rather than punitive function. It allows those evaluating academic work to understand the relationship between the submitted work and the technologies used during its production.

Importantly, disclosure does not imply that AI use is inappropriate. On the contrary, transparent acknowledgement often demonstrates professional responsibility. Just as researchers routinely disclose funding sources, conflicts of interest, or statistical software, AI use may represent another aspect of scholarly transparency rather than evidence of misconduct.

The educational value of disclosure therefore lies in making evaluation possible.

Disclosure Is Not Sufficient

Consider Emily's dissertation.

Suppose Emily openly declares:

"Artificial intelligence was used to generate interpretations of interview findings and to draft substantial parts of the discussion chapter."

Emily has behaved honestly.

She has complied with institutional expectations regarding transparency.

Yet an examiner may still conclude that AI assumed too much responsibility for interpreting the data.

The work may therefore remain academically problematic despite complete honesty.

Now consider a second declaration.

"Artificial intelligence was used to improve grammar, sentence clarity, and overall readability after the analytical content had been completed by the author."

Again, transparency has been maintained.

However, the educational judgement is likely to be very different because AI's function remained supportive rather than substitutive.

These examples illustrate a central principle of the framework:

Disclosure informs evaluation. It does not replace evaluation.

The Principle of Proportionate Disclosure

Not every interaction with AI requires identical levels of reporting.

Higher education already recognises proportionality in many other areas of academic practice. Minor editorial assistance is treated differently from statistical consultancy; language proofreading differs from collaborative authorship.

AI assistance should be approached similarly.

The extent of disclosure should broadly reflect the educational significance of AI's contribution.

Routine language refinement may require only minimal acknowledgement—or none at all where institutional policy explicitly permits this.

By contrast, AI involvement in idea generation, analytical interpretation, literature synthesis, methodological design, or research conclusions warrants substantially greater transparency because these activities directly influence scholarly judgement.

Proportionality therefore connects transparency to functional limitation. As AI's intellectual contribution increases, the expectation for meaningful disclosure becomes correspondingly stronger.

Transparency Protects Trust

Disclosure ultimately serves a broader purpose than institutional compliance.

It protects trust.

Academic qualifications retain value because employers trust that graduates possess the knowledge their degrees represent.

Research influences policy because readers trust that published conclusions genuinely reflect the authors' scholarly judgement.

Professional accreditation depends upon confidence that assessed competence belongs to the individual receiving recognition.

Whenever AI becomes involved in producing academic work, these relationships of trust require renewed attention.

Transparent disclosure allows trust to be maintained without requiring institutions to prohibit technology altogether.

Rather than concealing AI use, responsible scholars make it visible.

Rather than assuming transparency weakens credibility, the framework argues that appropriate disclosure strengthens it by demonstrating confidence in the legitimacy of the underlying intellectual work.

Transparency Without Responsibility Becomes Disclosure Theatre

One unintended consequence of current AI policies is the emergence of what might be called **disclosure theatre**.

Institutions sometimes devote considerable attention to declaration forms while paying relatively little attention to the quality of the intellectual work itself.

Students learn to complete disclosure statements.

Researchers add AI declarations to manuscripts.

Universities produce increasingly detailed reporting requirements.

Yet the fundamental educational question remains unanswered:

Who actually performed the thinking?

An institution that relies exclusively upon disclosure risks confusing administrative compliance with academic integrity.

Transparency is valuable precisely because it supports evaluation of responsibility and functional limitation.

Separated from those principles, disclosure risks becoming procedural rather than educational.

The framework therefore rejects both extremes.

Disclosure should neither be treated as evidence of misconduct nor accepted as sufficient evidence of legitimate authorship.

It functions as one essential component within a broader evaluative process.

Revisiting Emily

Emily discloses every use of AI honestly.

The framework does not automatically approve or reject her dissertation because disclosure alone cannot answer the central question.

Instead, disclosure enables examiners to ask more meaningful questions:

- Which interpretations originated with Emily?
- Which analytical decisions remained hers?
- Did AI support or replace disciplinary reasoning?
- Does the submitted dissertation still represent Emily's scholarly judgement?

Transparency therefore improves academic evaluation by making these questions visible rather than concealing them.

5. Academic Legitimacy as a Relational Outcome

Beyond Rule-Based Evaluation

Universities have traditionally governed academic integrity through rules. Students are instructed not to plagiarise, fabricate data, collude, or submit work produced by others. Such rules remain necessary because they define clear boundaries around unacceptable conduct.

Generative AI, however, exposes the limitations of purely rule-based governance.

Many forms of AI use cannot be classified simply as acceptable or unacceptable. They occupy a middle ground in which legitimacy depends upon context, educational purpose, disciplinary expectations, assessment design, and the continuing relationship between the submitted work and its human author.

Consider two students.

Both use AI to produce approximately one thousand words.

The first asks AI to rewrite sections of a draft they have already developed independently. AI improves expression but contributes no new ideas.

The second asks AI to generate the discussion chapter of an empirical report before making minor stylistic edits.

A rule based solely upon word count would treat both students similarly.

Professional academic judgement would not.

The difference lies not in the quantity of AI-generated text but in the relationship between AI and human intellectual work.

For this reason, the framework rejects binary models of AI governance in favour of **relational evaluation**.

Academic legitimacy is not a property of AI.

Nor is it a property of disclosure.

Nor is it determined solely by institutional policy.

Instead, legitimacy emerges through the interaction of responsibility, functional limitation, and transparency within the educational context in which the work is produced and evaluated.

Legitimacy Is Constructed Rather Than Declared

One of the assumptions challenged by generative AI is the belief that legitimacy can be declared administratively.

Institutions frequently ask students to sign declarations stating that submitted work is their own.

Researchers confirm authorship during journal submission.

Publishers require AI disclosure statements.

These declarations are important.

Yet declarations alone do not establish legitimacy.

Legitimacy is something that must be **constructed through evidence**.

Readers infer authorship from the quality of reasoning.

Supervisors infer learning from conversations with students.

Examiners infer understanding from assessment performance.

Editors infer scholarly responsibility from the coherence and originality of published work.

These are fundamentally evaluative activities.

The framework therefore treats legitimacy not as something granted by administrative procedures but as an interpretive judgement supported by multiple forms of evidence.

The Three Principles as an Integrated System

Each principle contributes something different to the evaluation of AI-assisted academic work.

Epistemic Responsibility answers the question:

Who owns the intellectual work?

Functional Limitation asks:

What role did AI actually perform?

Transparency and Disclosure asks:

Can evaluators understand how AI contributed to the work?

Only when all three questions receive satisfactory answers can academic legitimacy reasonably be sustained.

This interaction can be visualised as an evaluative system rather than a checklist.

If responsibility is absent, authorship disappears regardless of disclosure.

If AI performs intellectual work that should remain human, disclosure cannot restore legitimacy.

If AI use is concealed, evaluators cannot judge responsibility fairly.

The three principles therefore function collectively rather than sequentially.

Revisiting Emily

Emily's dissertation illustrates this interaction.

Responsibility

Emily independently designed the study, selected participants, analysed the interviews, rejected several AI suggestions, and defended every conclusion during her viva.

Responsibility appears well preserved.

Functional Limitation

However, AI generated much of the first draft of the discussion chapter and proposed comparisons with previous studies.

The examiner now asks:

Were these comparisons genuinely Emily's analytical contribution?

Or did AI perform disciplinary reasoning that the dissertation was intended to assess?

Transparency

Emily disclosed AI use fully.

This allows the examiner to evaluate the previous two questions openly rather than discovering AI involvement retrospectively.

Notice something important.

Emily's disclosure neither guarantees nor undermines legitimacy.

It simply enables a more informed judgement.

Likewise, AI's involvement does not automatically invalidate the dissertation.

The educational decision depends upon whether Emily retained meaningful intellectual ownership despite AI assistance.

This illustrates precisely why the framework is relational.

Legitimacy Exists Along a Continuum

Institutional discussions often imply that work is either legitimate or illegitimate.

Actual academic judgement is considerably more nuanced.

Between these extremes lie many situations requiring careful professional evaluation.

For example:

Situation	Likely judgement
AI corrects grammar after writing is complete	Strong legitimacy
AI improves structure while author retains ideas	Legitimate
AI suggests alternative explanations that the author critically evaluates	Generally legitimate but requiring judgement
AI develops substantial analytical interpretation	Increasing concern
AI generates conclusions that the author largely adopts	Weak legitimacy
AI performs the intellectual work while the human edits language	Academic legitimacy largely lost

The continuum is not intended to classify every possible case.

Rather, it reminds institutions that legitimacy often involves degrees of confidence rather than absolute certainty.

Professional judgement remains indispensable.

The Role of Evidence

Because legitimacy is relational, it cannot normally be established through a single indicator.

Instead, evaluators should consider multiple sources of evidence.

Possible indicators include:

- consistency between drafts;
- reflective commentary;
- oral explanation;
- research notebooks;
- revision history;
- engagement with sources;
- responses during supervision;
- ability to justify methodological choices;

- explanation of rejected alternatives;
- disciplinary reasoning beyond the submitted text.

None of these indicators independently proves authorship.

Together they create a richer picture of intellectual participation.

This represents an important shift.

Traditional academic integrity often focused on detecting misconduct.

The present framework focuses instead on identifying evidence of authentic intellectual engagement.

From Detection to Evaluation

Perhaps the most significant implication of the framework is the movement away from technological detection.

For several years, higher education invested considerable attention in AI detectors.

Experience has demonstrated their limitations.

False positives undermine trust.

False negatives encourage complacency.

More fundamentally, detection answers the wrong question.

Whether AI contributed to a piece of writing tells us little about whether meaningful learning occurred.

The framework therefore proposes a different orientation.

Instead of asking,

Can we detect AI?

educators should ask,

Can we evaluate authorship?

This subtle shift changes the purpose of assessment.

The objective becomes supporting authentic scholarship rather than policing technology.

Detection may still play a limited supporting role.

It should no longer define institutional thinking.

The Framework as a Professional Judgement Tool

The framework is not intended to replace academic judgement.

It is intended to improve it.

Professional judgement has always been central to higher education.

Markers interpret rubrics.

Supervisors evaluate dissertations.

Reviewers assess originality.

Editors judge contribution.

External examiners moderate standards.

None of these activities can be reduced to algorithms.

AI governance should be viewed similarly.

The framework therefore provides a shared language through which educators can discuss difficult cases more consistently while recognising that thoughtful disagreement will sometimes remain both inevitable and appropriate.

Transition to Chapter 6

Having established how legitimacy is evaluated, the remainder of the monograph shifts from theory to application.

The next chapter demonstrates how the framework can inform:

- assessment design;
- classroom practice;
- supervision;
- oral verification;
- moderation;
- institutional policy;
- quality assurance;
- academic integrity procedures.

The framework now moves from **explaining AI-assisted authorship** to **supporting institutional implementation**.

6. Applying the Framework in Higher Education

From Concept to Practice

Conceptual frameworks become valuable only when they influence practice. Universities require more than abstract principles; they need practical ways of applying those principles to assessment design, supervision, marking, moderation, and institutional governance. The purpose of this chapter is therefore not to introduce new theoretical ideas but to demonstrate how the framework can support everyday academic decision-making.

Importantly, the framework should not be viewed as another academic integrity policy. Policies define institutional expectations. The framework supports professional judgement when applying those expectations to increasingly complex cases of AI-assisted academic work.

Its value lies in helping educators ask better questions rather than providing automatic answers.

6.1 Assessment Design

Designing Assessments that Reward Thinking Rather than Text Production

Generative AI challenges a long-standing assumption within assessment: that producing written work necessarily demonstrates learning. While this assumption was never entirely accurate, AI makes its limitations more visible. Students can now generate coherent essays, reports, reflective pieces, and literature reviews with comparatively little intellectual effort.

The response should not be to eliminate written assessment. Academic writing remains central to higher education because it develops disciplinary thinking, communication, and scholarly identity. Instead, assessment should increasingly focus on making intellectual engagement visible throughout the learning process rather than evaluating only the final product.

This shift requires assessment designers to ask a different question.

Rather than asking:

"How can we prevent students from using AI?"

they should ask:

"What evidence would convince us that genuine intellectual engagement has occurred?"

This subtle change has significant implications for assessment design.

Assignments become more authentic when they require students to demonstrate activities that AI cannot easily perform independently, such as contextual judgement, personal reflection grounded in experience, discipline-specific interpretation, methodological justification, and critical evaluation of competing perspectives.

The framework therefore encourages assessment tasks that reward thinking rather than text production alone.

Principles for AI-Resilient Assessment

Assessment should be designed around learning outcomes rather than technology. Consequently, no assessment is completely AI-proof, nor should that be its objective. Instead, assessment should remain educationally meaningful regardless of whether AI is available.

The following principles support this objective.

Assess Intellectual Processes

Where possible, assess how students reached conclusions rather than evaluating conclusions alone.

Evidence may include:

- research planning;
- annotated reading;
- decision logs;
- methodological justification;
- reflective commentary;
- iterative drafting.

These activities reveal intellectual engagement that is often invisible within the final submission.

Require Contextual Application

Generic responses are easier for AI to generate than context-specific analysis.

Assessment tasks should therefore encourage students to apply disciplinary concepts to particular cases, local contexts, professional situations, or authentic datasets requiring human judgement.

The more context-dependent the task becomes, the greater the opportunity for students to demonstrate original disciplinary reasoning.

Value Critical Judgement

Many AI systems excel at summarising established knowledge.

Higher education, however, values judgement.

Students should therefore be expected to:

- evaluate evidence;
- justify decisions;
- defend interpretations;
- recognise uncertainty;
- compare competing explanations.

These activities make intellectual ownership substantially more visible than descriptive writing alone.

Build Assessment as a Process

Learning develops over time.

Assessment should therefore make important stages of intellectual development visible rather than relying exclusively upon final products.

Possible approaches include:

- proposal submission;
- annotated bibliography;
- draft feedback;

- progress meetings;
- oral defence;
- reflective commentary.

These components strengthen confidence that submitted work genuinely represents the student's learning journey.

6.2 Supervision

Supervising Learning Rather than Monitoring AI

Supervision occupies a distinctive position within higher education because it combines academic guidance with evaluation of intellectual development. Unlike traditional examinations, supervision allows educators to observe learning as it unfolds.

This places supervisors in a particularly strong position to evaluate AI-assisted work responsibly.

The framework encourages supervisors to move beyond questions such as:

"Did you use AI?"

towards more educationally valuable conversations.

Examples include:

- How did your argument develop?
- Which ideas changed during drafting?
- Why did you reject alternative explanations?
- Which sources most influenced your thinking?
- Where did AI assist you?
- Which AI suggestions did you decide not to use, and why?

These questions shift attention away from technology and towards scholarly judgement.

Students become accountable not merely for declaring AI use but for explaining their own intellectual decision-making.

Encouraging Responsible AI Literacy

Supervision should also recognise that responsible AI use forms part of contemporary academic literacy.

Students increasingly require guidance concerning:

- appropriate prompting;
- critical evaluation of AI outputs;
- recognising hallucinations;
- identifying bias;
- protecting confidentiality;
- maintaining authorship.

Supervisors therefore have responsibilities extending beyond misconduct prevention. They help students develop professional judgement regarding AI use that will remain valuable after graduation.

6.3 Marking and Feedback

Evaluating Intellectual Contribution

The framework does not require markers to determine exactly how much AI was used.

Instead, it encourages markers to evaluate whether sufficient evidence exists that the submitted work reflects meaningful intellectual engagement.

This changes the emphasis of marking.

Rather than searching for linguistic indicators of AI-generated writing, educators focus upon qualities traditionally associated with scholarship.

For example:

- conceptual coherence;
- disciplinary understanding;
- appropriate use of evidence;
- quality of analysis;
- consistency of reasoning;
- methodological justification;
- awareness of limitations.

These characteristics remain valuable regardless of technological developments.

Responding to Uncertainty

Not every concern regarding AI requires immediate misconduct investigation.

Where uncertainty exists, proportionate educational responses may be more appropriate.

Examples include:

- requesting clarification;
- discussing the work with the student;
- reviewing drafts;
- asking for additional explanation;
- requiring oral verification where justified.

Academic integrity procedures should remain available for serious concerns but should not become the default response whenever AI use is suspected.

The framework therefore encourages educational dialogue before disciplinary escalation whenever circumstances reasonably permit.

6.4 Moderation and Quality Assurance

Building Consistency Across Programmes

One of the greatest institutional challenges associated with AI lies not in technology itself but in inconsistency.

Students frequently encounter different expectations across modules, programmes, and departments.

One lecturer may encourage AI-assisted drafting.

Another may prohibit it entirely.

A third may require disclosure without explaining what should actually be disclosed.

Such inconsistency undermines fairness more than AI itself.

The framework therefore provides a shared conceptual language that moderation teams can use when discussing difficult cases.

Instead of debating individual AI tools, moderation discussions can consider common evaluative questions.

For example:

- Was intellectual responsibility preserved?
- Did AI remain functionally supportive?
- Was disclosure appropriate?
- Does sufficient evidence support authorship?

These shared questions encourage greater consistency without requiring identical disciplinary practices.

Supporting Institutional Learning

AI governance should itself become subject to continuous review.

Institutions should regularly examine:

- assessment outcomes;
- student experiences;
- staff confidence;
- emerging technologies;
- disciplinary developments;
- external expectations.

The framework is therefore intended as a living resource rather than a fixed regulatory model.

Universities should adapt implementation while preserving the underlying principles of responsibility, functional limitation, and transparency.

Chapter Summary

The framework supports higher education not by replacing professional judgement but by strengthening it.

Across assessment, supervision, marking, moderation, and quality assurance, the same three principles continue to operate.

Educators ask:

- Who remained intellectually responsible?

- What function did AI perform?
- Was AI use communicated transparently?

The answers may differ across disciplines and assessment contexts.

The evaluative questions remain remarkably consistent.

7. Relationship to the Companion Framework Suite

From Conceptual Framework to Institutional Practice

No single framework can address every challenge associated with AI-assisted academic work. Universities require conceptual clarity, but they also need practical tools that support consistent implementation across teaching, assessment, supervision, moderation, and quality assurance.

For this reason, *From Assistance to Authorship* should be understood as the conceptual foundation of a broader framework suite rather than as a standalone guidance document. It establishes the principles through which AI-assisted academic work should be evaluated. The companion resources translate those principles into practical institutional tools.

The relationship is intentional.

This proposed framework explains **why** legitimacy should be evaluated through responsibility, functional limitation, and transparency.

The companion documents explain **how** those principles can be applied within specific educational contexts.

Together they form an integrated approach to AI governance that moves beyond isolated policies towards a coherent institutional model.

The Framework Hierarchy

The framework suite consists of seven complementary documents.

1. *From Assistance to Authorship*

The conceptual foundation.

Purpose:

To define the principles that distinguish legitimate AI assistance from intellectual substitution.

Primary audience:

- researchers
- educators
- policy developers
- academic leaders

2. AI-Integrated Assessment Integrity Lifecycle

The institutional implementation model.

Purpose:

To demonstrate how responsible AI governance should be embedded across the complete assessment cycle, from assessment planning through quality enhancement.

Primary audience:

- programme leaders
- assessment designers
- quality assurance teams
- academic integrity units

3. AI Disclosure Framework

Purpose:

To support consistent, proportionate, and educationally meaningful disclosure of AI assistance by students and researchers.

Rather than encouraging administrative compliance alone, the framework promotes transparency as a mechanism for informed academic evaluation.

4. Oral Verification Toolkit

Purpose:

To provide structured guidance for confirming intellectual ownership where additional evidence of authorship is required.

Importantly, oral verification is presented as an educational conversation rather than an investigative procedure.

5. *Assessment Authenticity Checklist*

Purpose:

To support assessment design that encourages visible intellectual engagement while remaining compatible with responsible AI use.

The checklist assists educators in evaluating whether assessments continue to measure the intended learning outcomes within AI-supported learning environments.

6. *AI Risk-Level Matrix*

Purpose:

To help educators distinguish between supportive and substitutive forms of AI use by considering the educational significance of different AI functions.

Rather than classifying technologies themselves, the matrix classifies the educational risks associated with particular forms of AI contribution.

7. *Institutional AI Governance Guide for Assessment Integrity*

Purpose:

To support institutional implementation through policy development, staff development, moderation processes, quality assurance, and continuous improvement.

This document provides operational guidance for embedding the conceptual framework within institutional practice.

One Framework, Multiple Applications

Although each companion resource serves a different function, they share a common evaluative language.

Every document returns to the same three questions.

Responsibility

Who remained intellectually responsible?

Functional Limitation

What role did AI actually perform?

Transparency

Can that relationship be evaluated openly?

Maintaining consistent terminology across the suite is important.

Different institutional documents often introduce overlapping concepts using inconsistent language. As a result, staff may struggle to understand how disclosure guidance relates to assessment policy, or how oral verification connects to moderation procedures.

The present framework suite avoids this problem by establishing one conceptual vocabulary that underpins every companion resource.

Supporting Institutional Maturity

Many universities remain in the early stages of AI governance.

Initial responses have understandably focused on policy statements, academic integrity regulations, and student guidance. Over time, however, institutions are likely to require more mature governance arrangements.

Such maturity is characterised by several developments.

Institutions move:

- from prohibition towards responsible integration;
- from technology-centred policy towards learning-centred governance;
- from detecting AI towards evaluating authorship;
- from isolated procedures towards institution-wide coherence.

The framework suite has been designed to support this transition.

Rather than offering temporary guidance for current AI tools, it provides principles capable of adapting to future technological developments while preserving enduring academic values.

8. Scope and Limitations

Every conceptual framework necessarily reflects particular assumptions, priorities, and intended purposes. The present framework is no exception.

First, it is **normative** rather than predictive. It proposes how AI-assisted academic work should be evaluated; it does not attempt to predict how every educator or institution will make such judgements.

Second, the framework is **technology-neutral**. It does not distinguish between individual AI systems or recommend particular software. Its concern is the educational function performed by AI rather than the technical characteristics of specific models.

Third, the framework is intended to support **professional judgement**, not replace it. Universities differ in their educational missions, disciplinary traditions, assessment practices, and regulatory environments. Consequently, legitimate institutional variation should be expected. The framework seeks to improve consistency of reasoning rather than impose uniform decisions.

Fourth, while informed by scholarship, professional practice, and exploratory empirical research into academic perceptions of AI-assisted writing, the framework **has not yet undergone formal empirical validation as an assessment instrument**. It should therefore be interpreted as a conceptual and practice-informed framework rather than a validated measurement model.

Finally, the framework focuses primarily on **higher education**. Although many of its principles may transfer to professional training, secondary education, research publishing, or workplace learning, those contexts involve different purposes and governance requirements that merit independent consideration.

Recognising these limitations strengthens rather than weakens the framework. They clarify its intended contribution while identifying priorities for future research and institutional evaluation.

9. Future Development

Generative AI will continue to evolve. Consequently, AI governance should also evolve.

Future work should examine several areas.

Empirical research is needed to evaluate how consistently educators apply the framework across disciplines, institutions, and assessment contexts. Such studies may help refine operational guidance while identifying areas where additional clarification is required.

Assessment research should explore whether the framework supports more reliable moderation decisions, stronger assessment design, and improved student understanding of responsible AI use.

Institutional case studies may investigate how the companion framework suite influences policy development, academic integrity procedures, and quality assurance processes over time.

Finally, future scholarship should examine the relationship between AI-assisted authorship and broader questions concerning academic identity, disciplinary expertise, creativity, collaboration, and professional judgement. These issues extend beyond technology itself and touch the future purposes of higher education.

The framework presented here is therefore intended not as the conclusion of the conversation but as a starting point for continued scholarly discussion and institutional learning.

10. Conclusion

Generative artificial intelligence has transformed the conditions under which academic work is produced, evaluated, and trusted. While institutional responses have frequently concentrated on detection technologies, disclosure requirements, and lists of permitted or prohibited practices, these approaches do not fully address the central educational question:

When does AI-assisted work remain genuinely the work of its human author?

This proposed model has argued that the answer lies not in the presence or absence of AI, but in the preservation of three interdependent conditions.

The human author must retain **epistemic responsibility** for the intellectual content of the work.

AI must remain subject to **functional limitation**, supporting rather than replacing scholarly judgement.

Its contribution must be made visible through **transparency and disclosure**, allowing informed academic evaluation.

Together these principles reposition academic legitimacy as a relational judgement rather than a technological classification.

The framework therefore invites higher education to move beyond attempts to identify AI-generated text towards a more fundamental concern: preserving meaningful authorship within increasingly AI-mediated learning environments.

The question facing universities is no longer whether AI should exist within higher education. It already does.

The more important challenge is ensuring that technological assistance strengthens rather than weakens the intellectual responsibility upon which higher education ultimately depends.

Author's Note

This framework reflects an ongoing programme of research examining AI-assisted academic writing, assessment, and academic integrity within higher education. It draws upon published conceptual work, exploratory empirical research, and continuing professional engagement with assessment design and quality assurance. The framework should be understood as a conceptual, practice-informed contribution intended to stimulate scholarly discussion and support institutional adaptation rather than as a completed or universally validated governance model.

11. Version History

This publication is currently at Version 1.0, its initial release. Substantive revisions to the conceptual principles, the companion framework suite, or the empirical status of the framework will be recorded in future updates to this section, along with the corresponding version number and date of revision.

Appendix A

Figure 1: The From Assistance to Authorship Framework

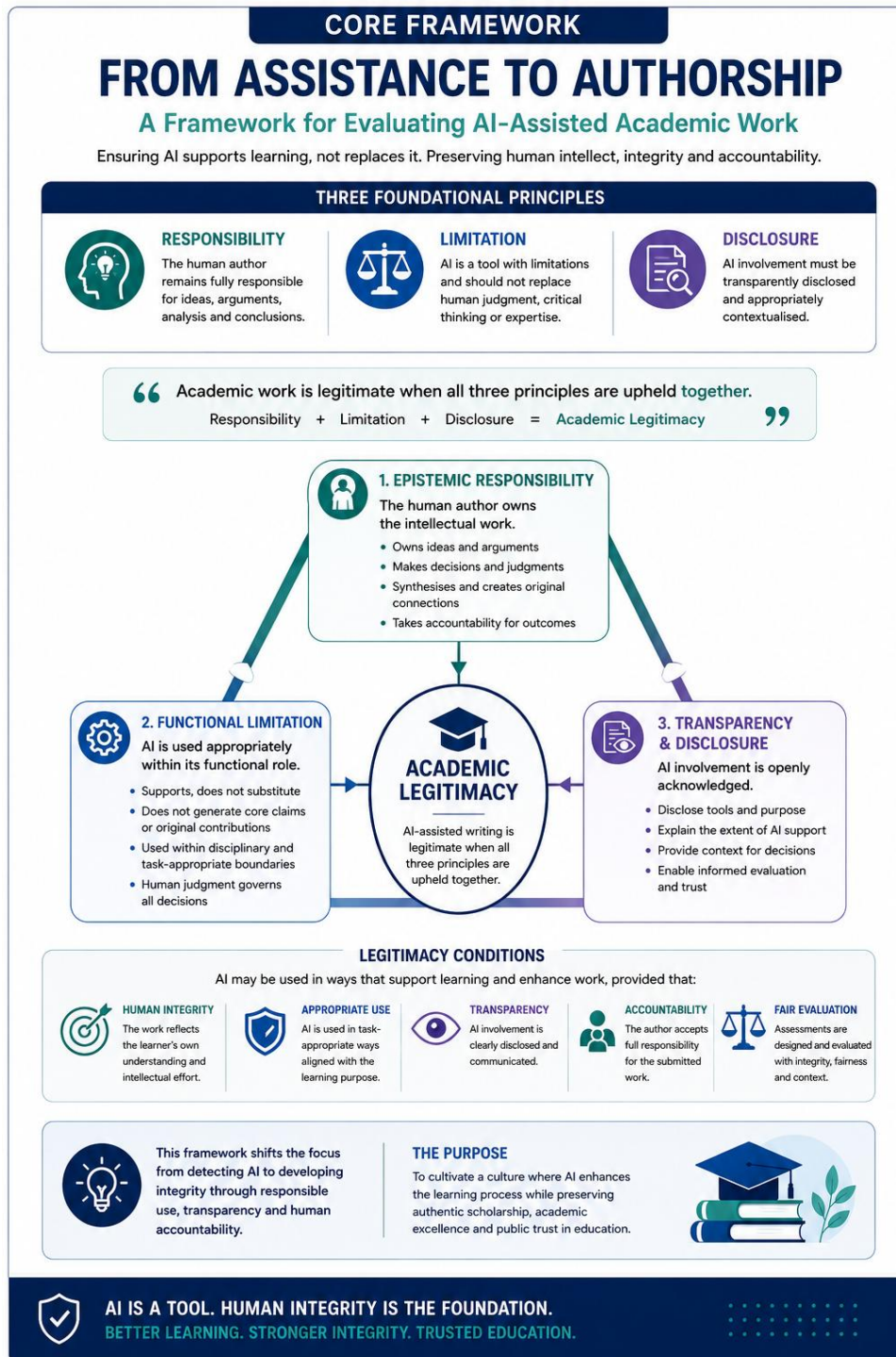


Figure 1. From Assistance to Authorship: A relational framework for evaluating AI-assisted academic work.

The framework conceptualises academic legitimacy as an evaluative outcome emerging from the interaction of three interdependent conditions: **Epistemic Responsibility**, **Functional Limitation**, and **Transparency and Disclosure**. Rather than classifying AI use as inherently acceptable or unacceptable, the framework proposes that legitimacy depends upon whether meaningful intellectual ownership remains attributable to the human author.

Table 1

The Three Foundational Principles

Principle	Central Question	Purpose
Epistemic Responsibility	Who performed the intellectual work?	Preserves authorship and scholarly accountability
Functional Limitation	What role did AI perform?	Prevents AI from replacing disciplinary reasoning
Transparency & Disclosure	Can evaluators understand AI involvement?	Supports informed academic judgement

Table 2

Supportive versus Substitutive AI Functions

Generally Supportive	Increasingly Substitutive
Grammar correction	Argument generation
Language refinement	Interpretation of findings
Formatting	Critical discussion
Reference formatting	Literature synthesis without evaluation
Translation	Drawing conclusions
Structural editing	Developing theoretical position
Accessibility support	Producing assessment responses demonstrating learning

Note. Classification depends upon educational purpose. Activities may be interpreted differently where the assessment specifically evaluates the skill being supported.

Table 3

Applying the Framework

Educational Activity	Responsibility	Functional Limitation	Transparency
Essay	Student develops argument	AI supports writing	AI declared where required
Dissertation	Student analyses evidence	AI supports drafting	Significant AI use disclosed
Laboratory Report	Student interprets results	AI edits language	Appropriate declaration
Literature Review	Student evaluates literature	AI supports organisation	AI involvement transparent
Reflective Writing	Student provides reflection	AI improves expression only	Institutional expectations followed

Appendix B

Illustrative Evaluation Questions

Epistemic Responsibility

- Who generated the central ideas?
- Who interpreted the evidence?
- Who decided which arguments to include?
- Can the author justify every major conclusion?
- Is disciplinary judgement clearly attributable to the human author?

Functional Limitation

- Did AI support expression or replace thinking?
- Which intellectual activities remained human?
- Would the learning outcome still have been demonstrated without AI performing analytical work?

- Did AI contribute to disciplinary reasoning?

Transparency

- Has AI involvement been disclosed?
- Is the disclosure proportionate?
- Does it allow evaluators to understand AI's contribution?
- Would reasonable readers be misled?

Appendix C

Emily Revisited

Principle	Evaluation
Responsibility	Emily remained responsible for research design and interpretation.
Functional Limitation	AI drafting of the discussion chapter requires closer examination because it approaches disciplinary reasoning.
Transparency	Full disclosure enables informed academic evaluation.
Overall Judgement	Further evidence required before confirming legitimacy.

This appendix demonstrates that the framework does not produce automatic decisions but structures professional judgement.

Appendix D

Intended Users

This framework has been developed for:

- University lecturers
- Academic developers
- Programme leaders
- Assessment designers
- External examiners
- Quality assurance professionals
- Academic integrity officers

- Educational researchers
- Policy developers
- Higher education leaders

Appendix E

Framework Status

Framework type

Conceptual framework

Development status

Practice-informed

Empirical status

The conceptual principles are informed by scholarly literature, professional practice, and exploratory empirical research examining academic judgements of AI-assisted writing. The framework itself has **not** been formally validated as an assessment instrument and should not be interpreted as a validated measurement model. The framework should therefore be regarded as a conceptual and practice-informed contribution that provides a foundation for future empirical validation and institutional evaluation.